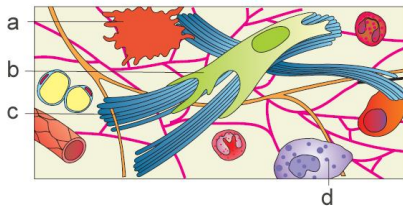


# STRUCTURAL ORGANIZATION IN ANIMALS

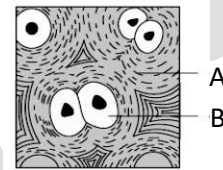
## ANIMAL TISSUES

- Which one is not suitable for epithelium?
  - It covers or lines body or body parts
  - Fibroblasts are present that release collagen
  - Compactly packed cells
  - It has free surface
- The epithelium found in the walls of blood vessels and alveoli is
  - Simple squamous
  - Simple cuboidal
  - Simple columnar
  - Glandular epithelium
- In animals, gametes are derived from
  - Epithelial tissue
  - Connective tissue
  - Neural tissue
  - Muscle tissue
- The type of tissue lining the nasal passage, bronchioles and fallopian tubes is
  - Cuboidal epithelium
  - Columnar ciliated epithelium
  - Germinal epithelium
  - Stratified columnar epithelium
- Find out the wrongly matched pair
  - Squamous epithelium – skin of frog
  - Columnar epithelium – peritoneum of body cavity
  - Ciliated epithelium – bronchioles
  - Stratified cuboidal epithelium – esophagus
  - Glandular epithelium – salivary gland
- Collagen and elastin are synthesized by
  - Fibroblasts
  - Macrophages
  - Mast cells
  - Adipocytes
- Given below is the diagrammatic sketch of a type of connective tissue. Identify the parts A, B, C & D.



- a. Macrophage      b. Fibroblast
  - c. Collagen fibres      d. Mast cell
  - a. Mast cell      b. Macrophage
  - c. Fibroblast      d. Collagen fibres
  - a. Macrophage      b. Collagen fibres
  - c. Fibroblast      d. Mast cell
  - a. Mast cell      b. Collagen fibres
  - c. Fibroblast      d. Macrophage
- Tendons
    - Attach skeletal muscles to bones
    - Attach one bone to another

- Attach muscle to muscle
  - All the above
- Cartilage cell is also known as
    - Osteocyte
    - Chondrocyte
    - Adipocyte
    - Hepatocyte
  - Which of the following is not a function of bones?
    - Protection
    - Hearing
    - Locomotion
    - Blood formation
    - Lymph formation
  - Analyze the following figure and select the correct statement.



- The figure is the section of bone and A is collagen fibres and B is osteocyte
  - The figure is the section of cartilage and A is collagen fibres and B is chondrocyte
  - The figure is the section of bone and A is collagen fibres and B is chondrocyte
  - The figure is the section of cartilage and A is collagen fibres and B is osteocyte
- Which of the following is fluid connective tissue?
    - Blood
    - Lymph
    - Saliva
    - Both A & B
  - Which is not true about skeletal muscle?
    - Dark and light bands are present
    - It has branched muscle fibres
    - Biceps muscle is an example
    - Myoglobin is present
  - Which one of the following feature is common to cardiac muscle and visceral muscle?
    - Both have uninucleate muscle fibres
    - Both have more mitochondria
    - Both have fibres with light & dark bands
    - Both have fusiform cells
  - Intercalated discs are present in
    - Striated muscle
    - Non-striated muscle
    - Cardiac muscle
    - All of these
  - Neural tissue is formed of
    - Brain, spinal cord and nerves
    - Neuron & capillaries
    - Cyton, axon and Dendron
    - Neuron & Neuroglia

## EARTHWORM: MORPHOLOGY & ANATOMY

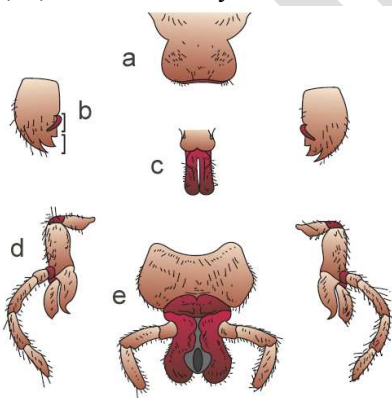
- The first segment of earthworm is known as
  - Clitellum
  - Prostomium
  - Peristomium
  - Typhlosole
- The function of calciferous glands in earthworm is to

- A. Lower blood Ca level  
B. Rise blood Ca level  
C. Neutralize humic acid in humus  
D. None of these
19. Which part of alimentary canal of earthworm is meant for grinding of the soil particles and decaying leaves?  
A. Buccal cavity      B. Typhlosole  
C. Gizzard            D. Stomach
20. In the 4<sup>th</sup>, 5<sup>th</sup> & 6<sup>th</sup> segments of earthworm, lying above pharyngeal mass and connected with pharyngeal glands are found small, red coloured follicular bodies called  
A. Septal glands      B. Blood glands  
C. Clitellum            D. Intestinal caecae  
E. Nephridia
21. Which one is correct?  
A. Blood glands are present on 4<sup>th</sup>, 5<sup>th</sup> & 6<sup>th</sup> segments of Pheretima  
B. Pharyngeal nephridia are present in the 4<sup>th</sup>, 5<sup>th</sup> & 6<sup>th</sup> segments of Pheretima  
C. Clitellum is found on 14-16 segments of Pheretima  
D. A pair of male genital pores are present on 14<sup>th</sup> segment of Pheretima  
E. All the above are correct
22. Which is not correct about the blood of earthworm?

- A. Phagocytic cells are present  
B. Haemoglobin is present  
C. Haemoglobin is seen on RBC  
D. All are correct
23. Select the correct sequence of sperm conduction in earthworm  
A. Testes → testis sac → seminal vesicle → testis sac → spermiducal funnel → vasa deferentia → prostatic duct → male genital aperture  
B. Testes → testis sac → spermiducal funnel → testis sac → seminal vesicle → prostatic duct → vasa deferentia → male genital aperture  
C. Testes → seminal vesicle → testis sac → spermiducal funnel → testis sac → vasa deferentia → prostatic duct → male genital aperture  
D. Testes → vasa deferentia → testis sac → seminal vesicle → testis sac → prostatic duct → spermiducal funnel → male genital aperture
24. Female genital aperture of earthworm is  
A. Double, ventral and seen on 14<sup>th</sup> segment  
B. Single, dorsal and seen on 18<sup>th</sup> segment  
C. Double, dorsal and seen on 18<sup>th</sup> segment  
D. Single, ventral and seen on 14<sup>th</sup> segment

## COCKROACH: MORPHOLOGY & ANATOMY

25. The dorsal plate of each segment of cockroach is called  
A. Sclerites            B. Tergites  
C. Sternites           D. None of these
26. In cockroach, Tegmina is  
A. A mouth part      B. Mesothoracic wing  
C. Metathoracic wing   D. A part of midgut
27. Label A, B, C, D & E correctly



- A. a. Labrum    b. Mandible    c. Hypopharynx  
d. Maxilla    e. Labium
- B. a. Labrum    b. Labium    c. Hypopharynx  
d. Mandible    e. Maxilla
- C. a. Labium    b. Maxilla    c. Hypopharynx  
d. Mandible    e. Labrum
- D. a. Mandible    b. Labrum    c. Labium  
d. Maxilla    e. Hypopharynx
28. In cockroach, Malpighian tubules are present in between  
A. Oesophagus and crop    B. Foregut & midgut  
C. Midgut and hindgut    D. Rectum and anus
29. Which structure is present only in male cockroach?

- A. Anal cercus            B. Anal style  
C. Anal sphincter       D. All the above
30. The number of abdominal segments in male and female cockroach is  
A. 10, 10                B. 9, 10  
C. 10, 11               D. 9, 9
31. Hypopharynx of cockroach is similar to  
A. Teeth                B. Tongue  
C. Buccal cavity       D. Oesophagus
32. Excretory organ in cockroach is  
A. Malpighian corpuscle    B. Malpighian tubules  
C. Hepatic caecae           D. All of these
33. In cockroach, Alary muscles are the parts of  
A. Digestive system    B. Respiratory system  
C. Circulatory system   D. Excretory system
34. Male and female cockroaches can be identified by observing  
A. Anal style            B. Anal cerci  
C. Abdomen            D. Antenna  
E. Both A & C
35. The blood of cockroach is called  
A. Haemocyte           B. Haemolymph  
C. Haemocoel           D. None of these
36. The eggs of cockroach are enclosed in  
A. Cocoon               B. Ootheca  
C. Clitellum            D. Genital chamber
37. Select the correct sequence  
A. Testis → seminal vesicle → vas deferens → ejaculatory duct → male gonopore

- B. Seminal vesicle → testis → vas deferens → ejaculatory duct → male gonopore  
C. Testis → vas deferens → seminal vesicle → ejaculatory duct → male gonopore

- D. Vas deferens → seminal vesicle → Testis → ejaculatory duct → male gonopore  
38. Nymph of cockroach becomes adult by moulting  
A. 3 times                      B. 13 times  
C. 10 times                     D. 5 times

## FROG: MORPHOLOGY & ANATOMY

39. Which set is more correct about frog?  
A. Poikilotherms, ear pinna, 3-chamberd heart  
B. Homoiotherms, ear pinna, 3-chamberd heart  
C. Homoiotherms, tympanum, 3-chamberd heart  
D. Poikilotherms, tympanum, 3-chamberd heart
40. During summer frog undergoes for  
A. Reproduction              B. Hibernation  
C. Aestivation                 D. All
41. Croaking of frog is  
A. Hunger call                 B. Sex call for female  
C. Danger call                 D. Musical note
42. Nuptial pad is the feature of  
A. Male cockroach            B. Female cockroach  
C. Male frog                    D. Female frog
43. In female frog, cloaca receives  
A. Ureter, bidder's canal and rectum  
B. Ureter, oviduct and rectum  
C. Urino-genital duct, oviduct and rectum  
D. Urino-genital duct, bidder's canal and rectum
44. Respiratory organs of frog are  
A. Lungs, gills, skin and trachea  
B. Lungs, trachea, book gills and skin  
C. Lungs, gills, skin and buccal cavity  
D. Lungs, book lungs, skin and trachea
45. Number of cranial nerves seen in frog  
A. 8 pairs                        B. 9 pairs  
C. 10 pairs                      D. 12 pairs
46. Bidder's canal is related with  
A. Reproductive system of male frog  
B. Excretory system of male frog  
C. Reproductive system of cockroach  
D. Excretory system of cockroach

>> **Answer Key:** [Click Here](#)

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